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25 March 1968
SUPERSEDING
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MILITARY SPECIFICATION

INSULATION SLEEVING, THERMAL, TUBULAR, FLEXIBLE

1. SCOPE

1.1 Scope. This specification covers flexible braided or woven tubular thermal insulation sleeving intended primarily for covering heater ducts, exhaust pipes, and other tubes in vehicles (see 6.1).

1.2 Classification.

1.2.1 Compositions. Tubular thermal insulation sleeving shall be one of the following compositions (materials) (see 6.2):

Composition I - Asbestos mixture
Composition II - Glass fiber

1.2.2 Ranges. Temperature ranges shall be one of the following (see 6.2):

Range 1 - Up to 1000°F
Range 2 - Up to 750°F
Range 3 - Up to 350°F

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Federal
CCC-T-191 - Textile Test Methods.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.

FSC 5640

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DRAWINGS
Army
A7384104 - Insulation, Heater Exhaust.
B7400075 - Insulation, Heater Exhaust.
B7991518 - Insulation, Tube, Heater Exhaust.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction samples. Preproduction samples of the insulation sleeving to be furnished under this specification shall be produced prior to the manufacture of these items in production quantity. The preproduction samples shall be properly identified and forwarded to the place specified by the procuring activity (see 6.2) for preproduction inspection as specified in 4.2. Preproduction samples submitted by the contractor shall be fully representative of the insulation sleeving to be supplied from production facilities.

3.2 Materials. Materials shall be free of dirt, knots, lumps, or other foreign matter detrimental to appearance of the finished product or to effective performance in its intended use (see 6.1). Materials shall not support fungus growth.

3.2.1 Composition I. Materials shall be a mixture of fibrous asbestos with cotton fibers, or other fibers, such that the finished insulation sleeving will meet the requirements of this specification. The yarn shall contain not less than 75 percent (by dry weight) asbestos.

3.2.2 Composition II. Materials shall be glass fibers of such composition, fineness, fiber length, and other characteristics that, without addition of any materials not necessary to the processing, the finished product will satisfy the requirements of this specification.

3.3 Construction.

3.3.1 Composition I. Insulation sleeving of this composition shall be of conventionally braided or woven construction, with multiple concentric braids used only when necessary in processing to provide the required wall thickness (see 6.3).

3.3.2 Composition II. Insulation sleeving of this composition shall be of tightly woven construction, with multiple concentric tubes used only when necessary in processing to provide the required wall thickness (see 6.3).

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3.4 Sizing. Sizing or similar materials, if used, shall be the minimum amount necessary to the manufacturing processes.

3.5 Adhesives. Cements, bonding agents, or other binding materials, if used, shall be the minimum amount necessary to secure an inner concentric tube to an outer tube. Any adhesive or other applied materials or treatment of cut ends to prevent separation of strands or yarns in handling shall not interfere with application of the finished product or performance in its intended use, and shall not yield poisonous fumes, continuous smoke, or noxious vapors.

3.6 Flexibility. Insulation sleeving shall be as pliable as practicable in relation to diameter and wall thickness, shall flatten for rolling onto a reel without damage, and shall resist collapse or kinking during application.

3.7 Moisture content.

3.7.1 Hygroscopic moisture. When tested as specified in 4.5, finished insulation sleeving hygroscopic moisture content shall not exceed the following:

Composition I material	-	3 percent
Composition II material	-	1 percent

3.7.2 Combined water (Composition I material). When tested as specified in 4.6, the chemically combined water in the dry asbestos fiber in the yarn used in the manufacture of insulation sleeving shall be not less than 12 percent.

3.8 Resistance to high temperature. After exposure to the high temperature tests specified in 4.7, the specimens shall show no evidence of broken strands, breakdown of structure, or any other damage that will impair the performance in its intended use.

3.9 Dimensions. Dimensions shall conform to the applicable drawings.

3.10 Marking. Marking shall be in accordance with the applicable drawing and MIL-STD-130, and shall include the following information.

Manufacturer's identification
Part number
Composition
Inside diameter
Temperature range

3.11 Instructions. When bulk insulation sleeving is furnished by the contractor, he shall supply with each unit package or reel approved printed instructions for cutting and protecting ends against fraying or separation of yarn or plies.

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3.12 Workmanship. Finished insulation sleeving shall be reasonably free of nap, broken and uneven threads, yarn or strand irregularities, foreign material, unspecified sizing or filler, and any other defects or imperfections that may affect serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for assuring that components and materials are examined and tested in accordance with the requirements of the referenced documents to the extent specified herein.

4.2 Preproduction inspection. Preproduction samples of a length, part number and quantity as specified (see 6.2) shall be submitted for preproduction inspection. The preproduction samples shall be examined for conformance to the quality assurance provisions of this specification.

4.2.1 Failure. Failure of a preproduction sample to satisfy any of the requirements specified herein will be cause for refusal by the Government to conduct additional inspection until satisfactory corrective measures have been taken by the contractor.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all insulation sleeving of one size, part number, material, and temperature range, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for testing. Samples for quality conformance testing shall be selected in accordance with inspection level S-2 of MIL-STD-105.

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4.3.2 Quality conformance examination.

4.3.2.1 Acceptable quality level. Samples selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For the purposes of examination, defects shall be classified as specified in table I.

Table I. Classification of defects

Categories	Defects
Critical	None defined.
Major:	
101	Materials not as specified (see 3.2.1 or 3.2.2).
102	Construction not as specified (see 3.3.1 or 3.3.2).
103	Adhesives, excessive use, installation interference, improper type (see 3.5).
104	Flexibility, too rigid; insufficient resistance to collapse or kinking (see 3.6).
105	Dimensions, nonconformance (see 3.9).
106	Printed instructions not provided (see 3.11).
Minor:	
201	Sizing, excessive (see 3.4).
202	Marking not as specified, illegible (see 3.10).
203	Workmanship, defects or imperfections (see 3.12).

4.4 Testing. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in 4.5 through 4.7.

4.4.1 Failure. Failure of any test sample to pass any test shall be cause for rejection of the lot and subsequent lots until the faults revealed by the tests have been corrected.

4.5 Hygroscopic moisture. To ascertain conformance to 3.7.1, not fewer than 5 specimens of suitable size, preferably weighing about 5 grams each, shall be brought to a stabilized moisture equilibrium condition with respect to atmosphere having a relative humidity of 65 percent ± 2 percent and temperature of 70° ± 2°F, in accordance with CCC-T-191, and each specimen carefully weighed.

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Specimens shall then be dried to constant weight in an electric oven maintained at 221° to 230°F, cooled in a desiccator, and weighed. Loss of weight during oven drying, multiplied by 100 and divided by weight after the establishment of moisture stabilization, shall be the percent of hygroscopic moisture. Average shall be taken of five or more specimens.

4.6 Combined water. To ascertain conformance to 3.7.2, specimens as specified in 4.5 shall be oven dried as specified in 4.5 and dry weight of each specimen recorded. Specimens shall then be subjected to a temperature of 1470° to 1490°F in an electric oven, then cooled in a desiccator and weighed. The difference between the dry weight and the residual weight, multiplied by 100 and divided by the dry weight, shall be the percentage of combined water. Average shall be taken of five or more specimens.

4.7 Resistance to high temperatures. To ascertain conformance to 3.8, insulation sleeving, after careful inspection for appearance and flexibility, shall be cut to specimen lengths of at least 12 times the specified internal diameter. Each specimen shall be placed on the outside of a tubular metal mandrel having the same outside diameter as the specified internal diameter of the insulation sleeving, and while on this mandrel, shall be subjected for one hour to a temperature 50°F above the upper temperature limit specified for the range. After cooling to 70° ± 10°F, the center 10-inch portion of each specimen shall be torsionally flexed five times alternately clockwise and counterclockwise to an angle of 15 degrees each way, after which careful inspection shall be made for broken or partly broken yarn or strands or other failure.

4.8 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the thermal insulation sleeving shall be examined to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging packing, and marking shall be in accordance with applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Insulation sleeving covered by this specification is intended for covering heater exhaust ducts, heater ducts or tubes, and pipes carrying fluids at elevated temperatures, in order to reduce heat losses and to protect personnel and property from injury and damage.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

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(b) Composition and temperature range (see 1.2).
(c) Preproduction samples, length and quantity (see 4.2).
(d) Place to send preproduction samples (see 3.1).
(e) Selection of applicable packaging standard or packaging data sheet (see 5.1).

6.3 Concentric tubes. When necessary, thick walls may be obtained by use of two or more concentric tubes, and suitable cement may be used to secure an inner tube to an outer tube, provided the resulting double-wall or multiple-wall insulation sleeving satisfies requirements of this specification (see 3.3.1 and 3.3.2).

Custodian:
Army - AT

Review activity:
Army - SM

Preparing activity:
Army -AT

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